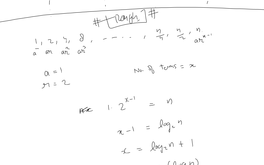
```
void fun(int n, bool b) {
    ...
    return fun(n/2, true) + fun(n/2, false);
}
```

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{2}\right) + c$$

$$T(n) = 2\,T\left(\frac{n}{2}\right) + c$$



[illegible]

$1, 2, 4, 8, \ldots, \frac{n}{2}, \frac{n}{4}, \frac{n}{8} \ldots$

$2^0 \quad 2^1 \quad 2^2 \quad 2^3$

$a = 1$

$r = 2$

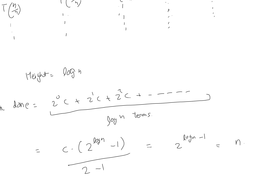
No of terms $= x$

$$1 \cdot 2^{x-1} = n$$

$$x - 1 = \log_2 n$$

$$x = \log_2 n + 1$$

$$x = O(\log n)$$



Height $= \log n$

work done $= \underbrace{2^0 c + 2^1 c + 2^2 c + \cdots}_{\log n \text{ terms}}$

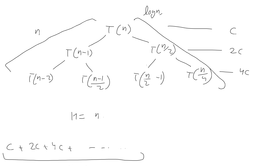
$$= \frac{c \cdot (2^{\log n} - 1)}{2 - 1} = 2^{\log n - 1} = n$$

work done $= O(n)$.

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{2}\right) + c$$

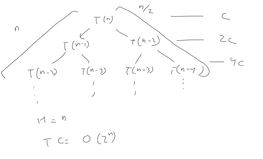
Example-2: F(n) = F(n/2) + F(n/4) + c

```
void fun(int n) {
    ...
    return fun(n/2) + fun(n/4);
}
```



$TC = O(n)$.

$$T(n) = T(n-1) + T\left(\frac{n}{2}\right) + c$$

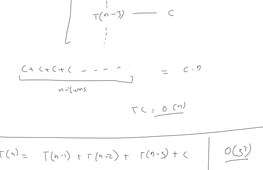


$H = n$

$$\underbrace{c + 2c + 4c + \cdots}_{n\text{-terms}}$$

$$c\,\frac{(2^n - 1)}{2 - 1} = O(2^n)$$

$$T(n) = T(n-1) + T(n-2) + c$$



$H = n$

$TC = O(2^n)$

$$T(n) = T(n-1) + c$$

$H = n$

$T(n) — c$

$T(n-1) — c$

$T(n-2) — c$

$T(n-3) — c$

$\vdots$

$$\underbrace{c + c + c + c + \cdots}_{n\text{-terms}} = c \cdot n$$

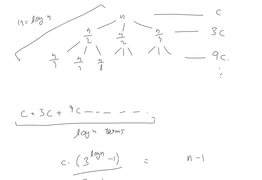
$TC = O(n)$

$$T(n) = T(n-1) + T(n-2) + T(n-3) + c$$

$$T(n) = 2\,T\left(\frac{n}{2}\right) + T\left(\frac{n}{3}\right) + c$$

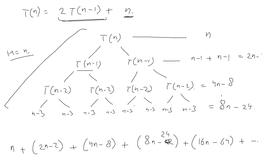
$O(3^n)$

$O(n)$



$$\underbrace{c + 3c + 9c + \cdots}_{n\text{-terms}}$$

$$c \cdot \frac{(3^n - 1)}{3 - 1} = O(3^n)$$



$$\underbrace{c + 3c + 9c + \cdots}_{\log n \text{ terms}}$$

$$c \cdot \frac{(3^{\log n} - 1)}{3 - 1} = n - 1$$

$$TC = O(n)$$

$$\boxed{\begin{aligned} 2^x &= n \\ x &= \log n \end{aligned}} \qquad \boxed{2^{\log n} = n}$$

$$T(n) = 2\,T(n-1) + n$$

$$n + (2n-2) + (4n-8) + (8n - 16) + (16n - 64) + \cdots$$

$$\underbrace{(n + 2n + 4n + 8n + \cdots)} - \underbrace{(2 + 8 + 24 + 64 + \cdots)}$$

$$(2 \cdot 1 + 4 \cdot 2 + 8 \cdot 3 + 16 \cdot 4 + 32 \cdot 5 \cdots)$$

$$n\,(1 + 2 + 4 + 8 + \cdots)$$

$$n\left(\frac{1 \cdot 2^n - 1}{2 - 1}\right) = n \cdot 2^n$$

$$= n \cdot 2^n$$

$$(n-1)\,2^{n+1} + 2$$

$$n \cdot 2^n - 2\,n\,2^{n+1} + 2^{n+1} - 2$$

$$2^{n+1} - n\,(2) - 2$$

$$=$$